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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B. Tech (CBCS) Degree Examinations

December 2025

CV2006-1 – GEOTECHNICAL ENGINEERING

(For CV)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. The dry unit weight is given by the equation
A) $G/(1-e)$ multiplied by unit weight of water
B) $(G+1)/(1-e)$ multiplied by unit weight of water
C) $G/(1+e)$ multiplied by unit weight of water
D) $(G-1)/(1-e)$ multiplied by unit weight of water
2. In a saturated soil, water content is 20% and specific gravity is 2.70. Its void ratio is
A) 0.54
B) 0.53
C) 0.52
D) 0.55
3. A soil sample has a field density of 16.5 kN/m^3 with a water content of 15%, its dry density is
A) 14.45 kN/m^3
B) 14.35 kN/m^3
C) 14.55 kN/m^3
D) 15.35 kN/m^3
4. The Uniformity coefficient is given by the expression
A) D_{10}/D_{60}
B) D_{60}/D_{10}
C) D_{30}/D_{10}
D) D_{10}/D_{30}
5. The liquid limit of soil is determined by.
A) Pycnometer
B) Density bottle
C) Casagrande's apparatus
D) Core cutter
6. In ISSCS GW stands for
A) Poorly graded gravel
B) Well graded sand
C) Poorly graded sand
D) Well graded gravel
7. The presence of stratification in soil layers affects permeability by:
A) Decreasing it
B) Increasing it
C) Having no effect
D) Making it uniform
8. The hydraulic gradient in a soil mass is defined as the ratio of:
A) Flow rate to area
B) Head loss to length of flow path
C) Permeability to porosity
D) Head loss to flow rate
9. In the Standard Proctor test, the soil is compacted in _____ layers.
A) 5
B) 4
C) 3
D) 2
10. The type of compaction equipment best suited for granular soils is the _____.
A) Smooth drum roller
B) Sheepfoot roller
C) Pneumatic roller
D) Vibratory roller
11. The _____ of the soil significantly affects its compaction characteristics.
A) Shear strength
B) Grain size distribution and Moisture content
C) Cohesion
D) Permeability
12. The pre-consolidation pressure is the maximum past _____ that a soil has experienced.
A) Effective stress
B) Void ratio
C) Water content
D) Total stress

13. In Terzaghi's one-dimensional consolidation theory, it is assumed that the soil is _____.
- A) Dry B) Partially saturated
C) Saturated D) Unsaturated
14. The consolidation process in Terzaghi's theory is assumed to be _____ dependent.
- A) Pressure B) Time
C) Stress D) Strain
15. The coefficient of consolidation (C_v) can be determined using the square root of time method by identifying the point corresponding to _____ consolidation.
- A) 70% B) 50%
C) 80% D) 90%
16. The secondary consolidation settlement occurs after _____ has been completed.
- A) Initial compression B) Primary consolidation
C) Immediate settlement D) Ultimate settlement
17. Compaction increases the density of soil, which generally _____ its shear strength.
- A) Increases B) Decreases
C) Remains constant D) Varies randomly
18. Sensitivity of a soil is defined as the ratio of _____ strength to disturbed strength.
- A) Residual B) Peak
C) Undisturbed D) Effective
19. The unconfined compression test is primarily used for _____ soils.
- A) Cohesive B) Cohesionless
C) Granular D) Gravelly
20. The safe bearing capacity is the ratio of _____.
- A) Factor of safety/Ultimate bearing capacity B) Ultimate bearing capacity/Factor of safety
C) Factor of safety/Net Ultimate bearing capacity D) Net Ultimate bearing capacity/Factor of safety

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) Explain the procedure to determine field density by sand replacement method.
b) Classify soils based on the data provided in the following table. Write step by step procedure of soil classification.

Soil	Liquid limit (%)	Plastic limit (%)	% passing through 0.075 mm sieve	% passing through 4.75mm sieve	Uniformity coefficient	Coefficient of curvature
A	42	26	65	90	-	-
B	26	22	10	20	7	1.1

2. a) A 2 cubic centimeter of soil in the natural state weighs 3.10 g. If it weighs 2.70 g after drying and has a specific gravity of 2.6, determine void ratio, porosity, degree of saturation and water content of soil, as it existed in the natural state.
b) With a neat sketch describe single grained and flocculent structures as applied to the fine-grained soils.
3. a) Determine the liquid limit, flow index, plasticity index and toughness index of soil sample from the following data obtained while conducting liquid limit test. Plastic limit of soil sample was taken as 24%.

Number of blows (N)	45	32	20	12
Water content (%)	24	30	41	50

- b) Explain the classification of coarse grained soil as per Indian soil classification system.

Marks BT* CO* PO*

08 L*1 1 1

08 L3 2 2

08 L3 1 1,2

08 L1 2 1,2

08 L3 1 1

05 L3 2 1

Unit – II

4. a) Discuss any four factors affecting permeability of soil. 04 L1 3 1,2
 b) Discuss any two methods adopted during compaction of soil in field. 04 L2 3 1,2
 c) A saturated soil has a compression index of 0.25. Its void ratio at stress 100 kN/m² is 0.8 and its permeability is 4×10^{-6} cm/s. Compute i) The change in void ratio if the stress is increased to 200 kN/m² ii) Time required for 60% consolidation of same soil 2.0 m thick in field with two-way drainage. 08 L3 4 4
5. a) The following are the results of a standard compaction test performed on a sample of soil. Take the volume of mould used was 960 cc. Plot compaction curve and estimate optimum moisture content and maximum dry density.
- | | | | | | | |
|------------------------|------|------|------|------|------|------|
| Water content (%) | 9 | 11 | 14 | 17 | 20 | 23 |
| Weight of wet soil (g) | 1750 | 1950 | 2250 | 2200 | 2100 | 1950 |
- b) Explain i) Primary consolidation ii) Compression index. 08 L3 3 1,2
 c) Describe the concept of consolidation with the help of mass spring analogy. 04 L1 4 2
 04 L1 4 2
6. a) Discuss any four factors affecting compaction property of soil. 08 L1 3 1,2
 b) Explain in detail, how to determine the coefficient of consolidation by square root of time fitting method. 08 L1 4 3

Unit – III

7. a) A cylindrical specimen of a saturated soil fails under an axial stress of 160 kN/m² in an unconfined compression test. The failure plane makes an angle of 48° with the horizontal. Calculate the cohesion and angle of internal friction of the soil. 08 2 5 1
 b) A strip footing 1.2 m wide is supported on a dense soil with its base at a depth of 1.5 m below ground level. The soil properties are as follows; $c = 20$ kN/m², $\phi = 34^\circ$, $\gamma = 18$ kN/m³, $\gamma' = 10$ kN/m³. Determine the ultimate bearing capacity if i) water table is at great depth ii) water table is at the level of base of the foundation iii) water table is at the ground surface. Use Terzaghi's equation. Take $N_c = 52.6$, $N_q = 36.5$ and $N_\gamma = 35.0$. 08 3 5 3
8. a) The samples of compacted, clean dry sand were tested in a shear box 60mmx60mm and the following results were obtained.
- | | | | | |
|----------------|-----|-----|-----|-----|
| Normal load(N) | 101 | 203 | 299 | 401 |
| Shear Load (N) | 55 | 116 | 173 | 241 |
- Determine the angle of shearing resistance. Draw the Mohr circle for the second trial. Also mark the failure plane, major and minor principal planes. 08 2 5 1
 b) List the characteristics of general shear failure and local shear failure. 08 3 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
